SCSR2043 OPERATING SYSTEMS

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Section	:	01	

This lab assessment is designed to test your understanding and skills on some basic concepts and tools related to process monitoring and management in operating system. Please follow the instructions carefully and submit your answers in this word document and rename the file as **os-lab-assessment02-studentname-matricno.docx**.

Essential Steps Before Starting Lab Assessment 2:

1. **Download necessary source codes:**

   Use the `wget` command to retrieve the following source code files to your Linux (or WSL or MacOS) environment:

```
wget -O mainprocess.c https://rebrand.ly/mainprocess_c
wget -O subprocess1.c https://rebrand.ly/subprocess1_c
wget -O subprocess2.c https://rebrand.ly/subprocess2_c
```

2. **Compile the source files:**

   Use the `gcc` compiler to create executable files from the source code.

```
gcc mainprocess.c -o mainprocess
gcc subprocess1.c -o subprocess1
gcc subprocess2.c -o subprocess2
```

3. **Execute the dummy processes:**

   Run all the dummy processes

```
./mainprocess &
```

   Press **enter** two times.

4. The dummy processes are running for 2 hours. If you took longer than 2 hours on questions 1-9, please restart the main process with `./mainprocess &`.

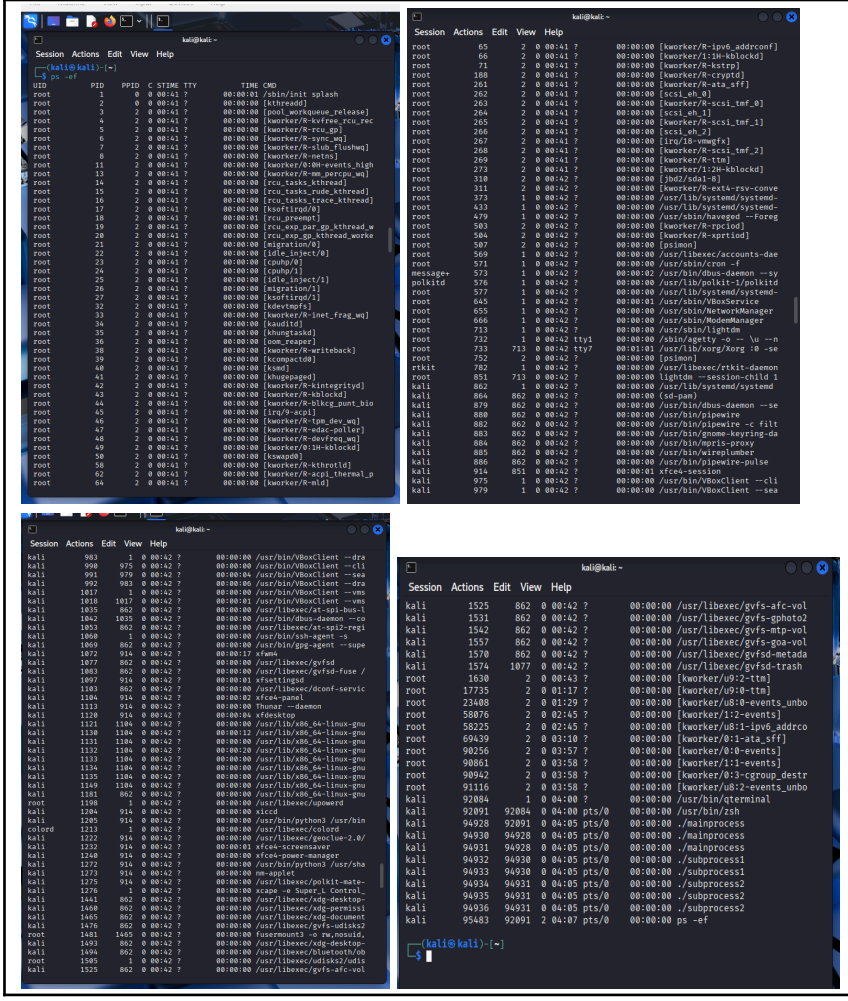
Lab Assessment 2 : Linux Process Monitoring and Management

Instructions:

1. Carefully execute each command as instructed in the questions.
2. Write down the exact command used for each task.
3. Capture a screenshot of the command's output.

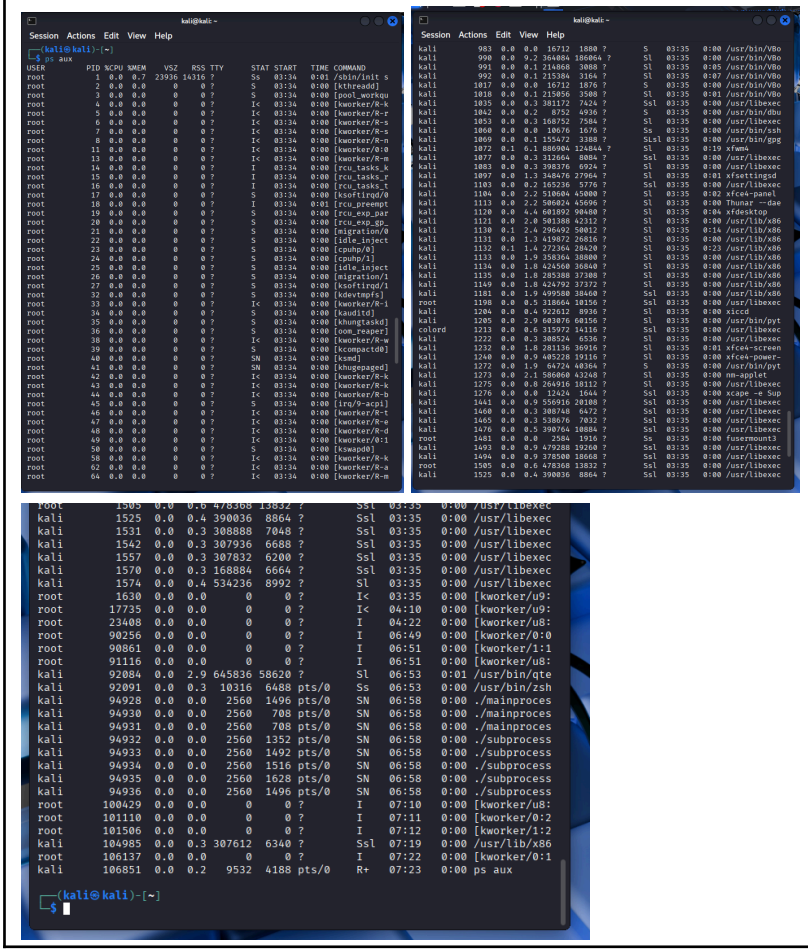
Question 1

Use the `ps` command with the appropriate option to display a complete list of all running processes within the Linux operating system.

Command
ps -ef
Output


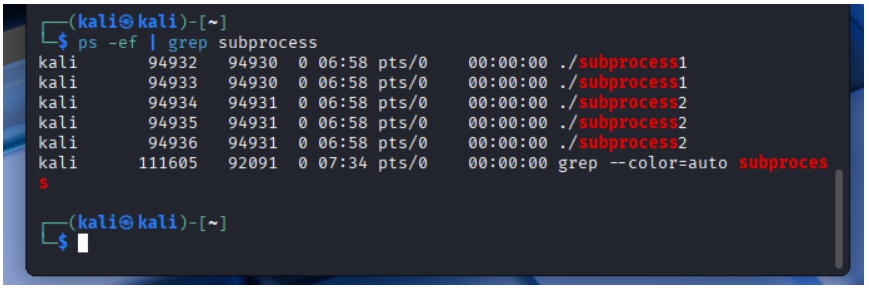
Question 2

Employ the `ps` command with necessary options to unveil comprehensive details about each running process.

Command
ps aux
Output


Question 3

Use the `ps` command with some tools to only list processes named "subprocess" and show some info about them.

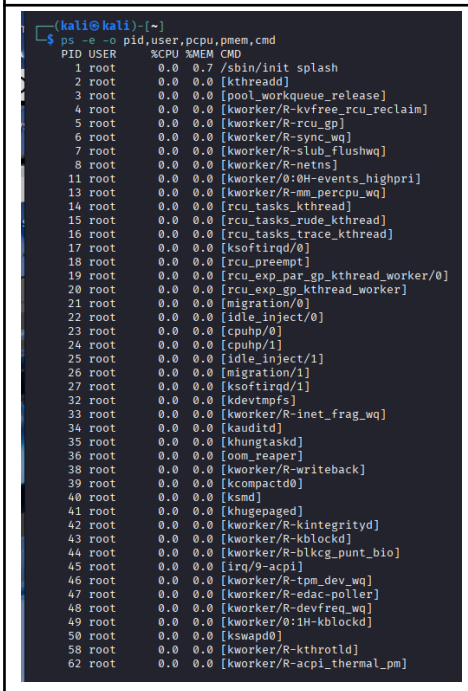
Command
ps -ef | grep subprocess
Output


Question 4

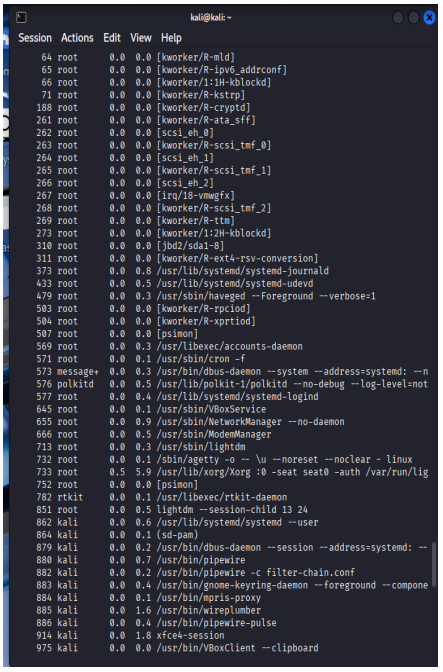
Execute the `ps` command, specifying options that reveal only the following columns:

- Process ID (`pid`)
- Owner of the process (`user`)
- CPU percentage (`pcpu`)
- Memory percentage (`pmem`)
- Command (`cmd`)

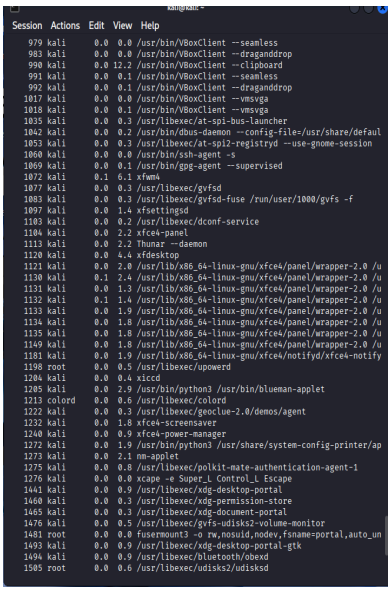
Command
ps -e -o pid,user,pcpu,pmem,cmd
Output



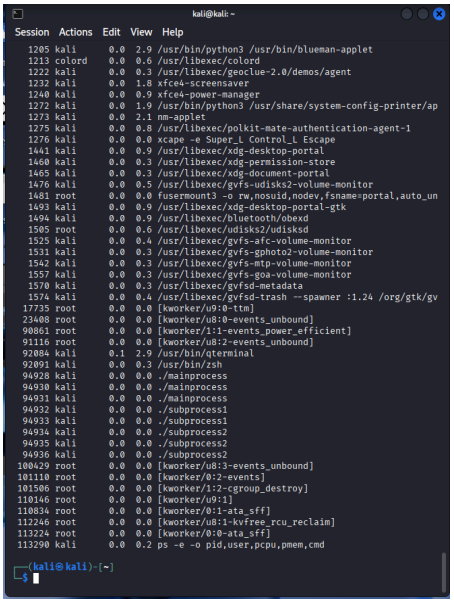
```
┌──(kali㉿kali)-[~]
└─$ ps -e -o pid,user,pcpu,pmem,cmd
    PID USER     %CPU %MEM CMD
      1 root      0.0  0.7 /sbin/init splash
      2 root      0.0  0.0 [kthreadd]
      3 root      0.0  0.0 [pool_workqueue_release]
      4 root      0.0  0.0 [kworker/R-kvfree_rcu_reclaim]
      5 root      0.0  0.0 [kworker/R-rcu_gp]
      6 root      0.0  0.0 [kworker/R-sync_wq]
      7 root      0.0  0.0 [kworker/R-slub_flushwq]
      8 root      0.0  0.0 [kworker/R-netns]
     11 root      0.0  0.0 [kworker/0:0H-events_highpri]
     13 root      0.0  0.0 [kworker/R-mm_percpu_wq]
     14 root      0.0  0.0 [rcu_tasks_kthread]
     15 root      0.0  0.0 [rcu_tasks_rude_kthread]
     16 root      0.0  0.0 [rcu_tasks_trace_kthread]
     17 root      0.0  0.0 [ksoftirqd/0]
     18 root      0.0  0.0 [rcu_preempt]
     19 root      0.0  0.0 [rcu_exp_par_gp_kthread_worker/0]
     20 root      0.0  0.0 [rcu_exp_gp_kthread_worker]
     21 root      0.0  0.0 [migration/0]
     22 root      0.0  0.0 [idle_inject/0]
     23 root      0.0  0.0 [cpuhp/0]
     24 root      0.0  0.0 [cpuhp/1]
     25 root      0.0  0.0 [idle_inject/1]
     26 root      0.0  0.0 [migration/1]
     27 root      0.0  0.0 [ksoftirqd/1]
     32 root      0.0  0.0 [kdevtmpfs]
     33 root      0.0  0.0 [kworker/R-inet_frag_wq]
     34 root      0.0  0.0 [kauditd]
     35 root      0.0  0.0 [khungtaskd]
     36 root      0.0  0.0 [oom_reaper]
     38 root      0.0  0.0 [kworker/R-writeback]
     39 root      0.0  0.0 [kcompactd0]
     40 root      0.0  0.0 [ksmd]
     41 root      0.0  0.0 [khugepaged]
     42 root      0.0  0.0 [kworker/R-kintegrityd]
     43 root      0.0  0.0 [kworker/R-kblockd]
     44 root      0.0  0.0 [kworker/R-blkcg_punt_bio]
     45 root      0.0  0.0 [irq/9-acpi]
     46 root      0.0  0.0 [kworker/R-tpm_dev_wq]
     47 root      0.0  0.0 [kworker/R-edac-poller]
     48 root      0.0  0.0 [kworker/R-devfreq_wq]
     49 root      0.0  0.0 [kworker/0:1H-kblockd]
     50 root      0.0  0.0 [kswapd0]
     58 root      0.0  0.0 [kworker/R-kthrotld]
     62 root      0.0  0.0 [kworker/R-acpi_thermal_pm]
```



```
kali@kali: ~
Session  Actions  Edit  View  Help
     64 root      0.0  0.0 [kworker/R-mld]
     65 root      0.0  0.0 [kworker/R-ipv6_addrconf]
     66 root      0.0  0.0 [kworker/1:1H-kblockd]
     71 root      0.0  0.0 [kworker/R-kstrp]
    188 root      0.0  0.0 [kworker/R-cryptd]
    261 root      0.0  0.0 [kworker/R-ata_sff]
    262 root      0.0  0.0 [scsi_eh_0]
    263 root      0.0  0.0 [kworker/R-scsi_tmf_0]
    264 root      0.0  0.0 [scsi_eh_1]
    265 root      0.0  0.0 [kworker/R-scsi_tmf_1]
    266 root      0.0  0.0 [scsi_eh_2]
    267 root      0.0  0.0 [irq/18-vmwgfx]
    268 root      0.0  0.0 [kworker/R-scsi_tmf_2]
    269 root      0.0  0.0 [kworker/R-ttm]
    273 root      0.0  0.0 [kworker/1:2H-kblockd]
    310 root      0.0  0.0 [jbd2/sda1-8]
    311 root      0.0  0.0 [kworker/R-ext4-rsv-conversion]
    373 root      0.0  0.8 /usr/lib/systemd/systemd-journald
    433 root      0.0  0.5 /usr/lib/systemd/systemd-udevd
    479 root      0.0  0.3 /usr/sbin/haveged --Foreground --verbose=1
    503 root      0.0  0.0 [kworker/R-rpciod]
    504 root      0.0  0.0 [kworker/R-xprtiod]
    507 root      0.0  0.0 [psimon]
    569 root      0.0  0.3 /usr/libexec/accounts-daemon
    571 root      0.0  0.1 /usr/sbin/cron -f
    573 message+  0.0  0.3 /usr/bin/dbus-daemon --system --address=systemd: --n
    576 polkitd   0.0  0.5 /usr/lib/polkit-1/polkitd --no-debug --log-level=not
    577 root      0.0  0.4 /usr/lib/systemd/systemd-logind
    645 root      0.0  0.1 /usr/sbin/VBoxService
    655 root      0.0  0.9 /usr/sbin/NetworkManager --no-daemon
    666 root      0.0  0.5 /usr/sbin/ModemManager
    713 root      0.0  0.3 /usr/sbin/lightdm
    732 root      0.0  0.1 /sbin/agetty -o -- \u --noreset --noclear - linux
    733 root      0.5  5.9 /usr/lib/xorg/Xorg :0 -seat seat0 -auth /var/run/lig
    752 root      0.0  0.0 [psimon]
    782 rtkit     0.0  0.1 /usr/libexec/rtkit-daemon
    851 root      0.0  0.5 lightdm --session-child 13 24
    862 kali      0.0  0.6 /usr/lib/systemd/systemd --user
    864 kali      0.0  0.1 (sd-pam)
    879 kali      0.0  0.2 /usr/bin/dbus-daemon --session --address=systemd: --
    880 kali      0.0  0.7 /usr/bin/pipewire
    882 kali      0.0  0.2 /usr/bin/pipewire -c filter-chain.conf
    883 kali      0.0  0.4 /usr/bin/gnome-keyring-daemon --foreground --compone
    884 kali      0.0  0.1 /usr/bin/mpris-proxy
    885 kali      0.0  1.6 /usr/bin/wireplumber
    886 kali      0.0  0.4 /usr/bin/pipewire-pulse
    914 kali      0.0  1.8 xfce4-session
    975 kali      0.0  0.0 /usr/bin/VBoxClient --clipboard
```



```
Session  Actions  Edit  View  Help
    979 kali      0.0  0.0 /usr/bin/VBoxClient --seamless
    983 kali      0.0  0.0 /usr/bin/VBoxClient --draganddrop
    990 kali      0.0 12.2 /usr/bin/VBoxClient --clipboard
    991 kali      0.0  0.1 /usr/bin/VBoxClient --seamless
    992 kali      0.0  0.1 /usr/bin/VBoxClient --draganddrop
   1017 kali      0.0  0.0 /usr/bin/VBoxClient --vmsvga
   1018 kali      0.0  0.1 /usr/bin/VBoxClient --vmsvga
   1035 kali      0.0  0.3 /usr/libexec/at-spi-bus-launcher
   1042 kali      0.0  0.2 /usr/bin/dbus-daemon --config-file=/usr/share/defaul
   1053 kali      0.0  0.3 /usr/libexec/at-spi2-registryd --use-gnome-session
   1060 kali      0.0  0.0 /usr/bin/ssh-agent -s
   1069 kali      0.0  0.1 /usr/bin/gpg-agent --supervised
   1072 kali      0.1  6.1 xfwm4
   1077 kali      0.0  0.3 /usr/libexec/gvfsd
   1083 kali      0.0  0.3 /usr/libexec/gvfsd-fuse /run/user/1000/gvfs -f
   1097 kali      0.0  1.4 xfsettingsd
   1103 kali      0.0  0.2 /usr/libexec/dconf-service
   1104 kali      0.0  2.2 xfce4-panel
   1113 kali      0.0  2.2 thunar --daemon
   1120 kali      0.0  4.4 xfdesktop
   1121 kali      0.0  2.0 /usr/lib/x86_64-linux-gnu/xfce4/panel/wrapper-2.0 /u
   1130 kali      0.1  2.4 /usr/lib/x86_64-linux-gnu/xfce4/panel/wrapper-2.0 /u
   1131 kali      0.0  1.3 /usr/lib/x86_64-linux-gnu/xfce4/panel/wrapper-2.0 /u
   1132 kali      0.1  1.4 /usr/lib/x86_64-linux-gnu/xfce4/panel/wrapper-2.0 /u
   1133 kali      0.0  1.9 /usr/lib/x86_64-linux-gnu/xfce4/panel/wrapper-2.0 /u
   1134 kali      0.0  1.8 /usr/lib/x86_64-linux-gnu/xfce4/panel/wrapper-2.0 /u
   1135 kali      0.0  1.8 /usr/lib/x86_64-linux-gnu/xfce4/panel/wrapper-2.0 /u
   1149 kali      0.0  1.8 /usr/lib/x86_64-linux-gnu/xfce4/panel/wrapper-2.0 /u
   1181 kali      0.0  1.9 /usr/lib/x86_64-linux-gnu/xfce4/notifyd/xfce4-notify
   1198 root      0.0  0.5 /usr/libexec/upowerd
   1204 kali      0.0  0.4 xiccd
   1205 kali      0.0  2.9 /usr/bin/python3 /usr/bin/blueman-applet
   1213 colord    0.0  0.6 /usr/libexec/colord
   1222 kali      0.0  0.3 /usr/libexec/geoclue-2.0/demos/agent
   1232 kali      0.0  1.8 xfce4-screensaver
   1240 kali      0.0  0.9 xfce4-power-manager
   1272 kali      0.0  1.9 /usr/bin/python3 /usr/share/system-config-printer/ap
   1273 kali      0.0  2.1 nm-applet
   1275 kali      0.0  0.8 /usr/libexec/polkit-mate-authentication-agent-1
   1276 kali      0.0  0.0 xcape -e Super_L Control_L Escape
   1441 kali      0.0  0.9 /usr/libexec/xdg-desktop-portal
   1460 kali      0.0  0.3 /usr/libexec/xdg-permission-store
   1465 kali      0.0  0.3 /usr/libexec/xdg-document-portal
   1476 kali      0.0  0.5 /usr/libexec/gvfs-udisks2-volume-monitor
   1481 root      0.0  0.0 fusermount3 -o rw,nosuid,nodev,fsname=portal,auto_un
   1493 kali      0.0  0.9 /usr/libexec/xdg-desktop-portal-gtk
   1494 kali      0.0  0.9 /usr/libexec/bluetooth/obexd
   1505 root      0.0  0.6 /usr/libexec/udisks2/udisksd
```



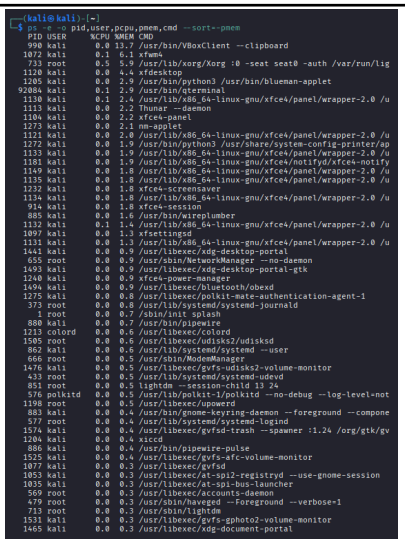
```
kali@kali: ~
Session  Actions  Edit  View  Help
   1205 kali      0.0  2.9 /usr/bin/python3 /usr/bin/blueman-applet
   1213 colord    0.0  0.6 /usr/libexec/colord
   1222 kali      0.0  0.3 /usr/libexec/geoclue-2.0/demos/agent
   1232 kali      0.0  1.8 xfce4-screensaver
   1240 kali      0.0  0.9 xfce4-power-manager
   1272 kali      0.0  1.9 /usr/bin/python3 /usr/share/system-config-printer/ap
   1273 kali      0.0  2.1 nm-applet
   1275 kali      0.0  0.8 /usr/libexec/polkit-mate-authentication-agent-1
   1276 kali      0.0  0.0 xcape -e Super_L Control_L Escape
   1441 kali      0.0  0.9 /usr/libexec/xdg-desktop-portal
   1460 kali      0.0  0.3 /usr/libexec/xdg-permission-store
   1465 kali      0.0  0.3 /usr/libexec/xdg-document-portal
   1476 kali      0.0  0.5 /usr/libexec/gvfs-udisks2-volume-monitor
   1481 root      0.0  0.0 fusermount3 -o rw,nosuid,nodev,fsname=portal,auto_un
   1493 kali      0.0  0.9 /usr/libexec/xdg-desktop-portal-gtk
   1494 kali      0.0  0.9 /usr/libexec/bluetooth/obexd
   1505 root      0.0  0.6 /usr/libexec/udisks2/udisksd
   1525 kali      0.0  0.4 /usr/libexec/gvfs-afc-volume-monitor
   1531 kali      0.0  0.3 /usr/libexec/gvfs-gphoto2-volume-monitor
   1542 kali      0.0  0.3 /usr/libexec/gvfs-mtp-volume-monitor
   1557 kali      0.0  0.3 /usr/libexec/gvfs-goa-volume-monitor
   1570 kali      0.0  0.3 /usr/libexec/gvfsd-metadata
   1574 kali      0.0  0.4 /usr/libexec/gvfsd-trash --spawner :1.24 /org/gtk/gv
  17735 root      0.0  0.0 [kworker/u9:0-ttm]
  23408 root      0.0  0.0 [kworker/u8:0-events_unbound]
  90861 root      0.0  0.0 [kworker/1:1-events_power_efficient]
  91116 root      0.0  0.0 [kworker/u8:2-events_unbound]
  92084 kali      0.1  2.9 /usr/bin/qterminal
  92091 kali      0.0  0.3 /usr/bin/zsh
  94928 kali      0.0  0.0 ./mainprocess
  94930 kali      0.0  0.0 ./mainprocess
  94931 kali      0.0  0.0 ./mainprocess
  94932 kali      0.0  0.0 ./subprocess1
  94933 kali      0.0  0.0 ./subprocess1
  94934 kali      0.0  0.0 ./subprocess2
  94935 kali      0.0  0.0 ./subprocess2
  94936 kali      0.0  0.0 ./subprocess2
 100429 root      0.0  0.0 [kworker/u8:3-events_unbound]
 101110 root      0.0  0.0 [kworker/0:2-events]
 101506 root      0.0  0.0 [kworker/1:2-cgroup_destroy]
 110146 root      0.0  0.0 [kworker/u9:1]
 110834 root      0.0  0.0 [kworker/0:1-ata_sff]
 112246 root      0.0  0.0 [kworker/u8:1-kvfree_rcu_reclaim]
 113224 root      0.0  0.0 [kworker/0:0-ata_sff]
 113290 kali      0.0  0.2 ps -e -o pid,user,pcpu,pmem,cmd

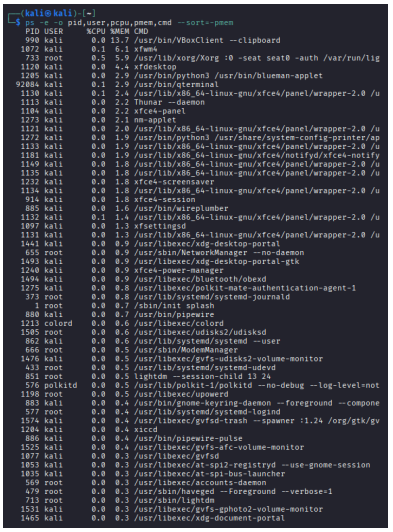
┌──(kali㉿kali)-[~]
└─$
```

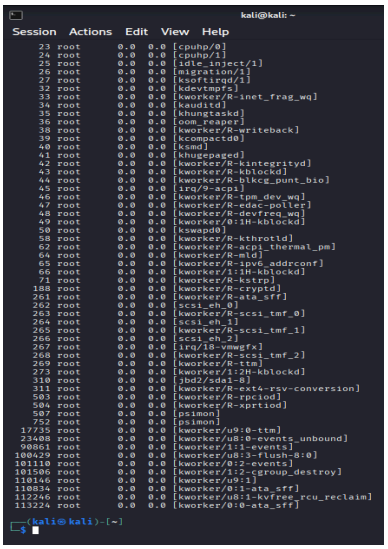
Question 5

Building on the ps command used in Question 4, can you add an option to sort the listed processes by their memory usage (`pmem`)?

Command
ps -e -o pid,user,pcpu,pmem,cmd --sort=-pmem
Output



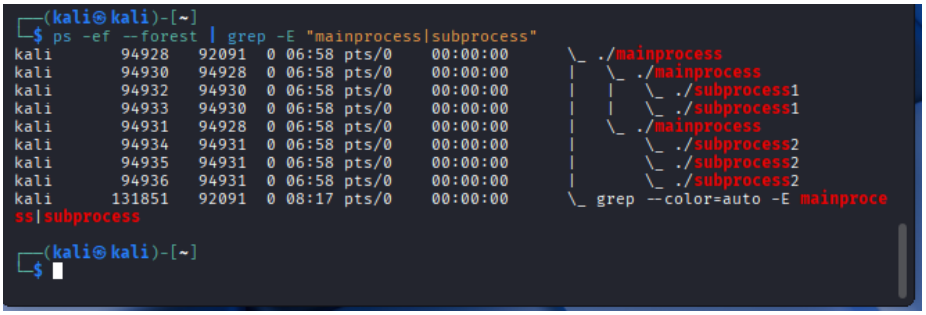




Question 6

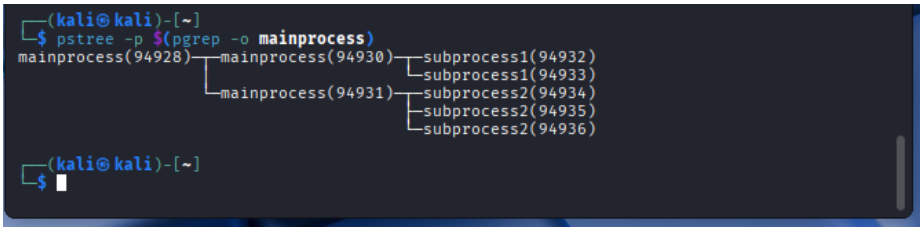
Construct a command using `ps`, suitable options, and any additional tools to visualize the hierarchical structure (tree-like) of the following processes:

- "mainprocess"
- "subprocess1"
- "subprocess2"

Command
ps -ef --forest | grep -E "mainprocess|subprocess"
Output


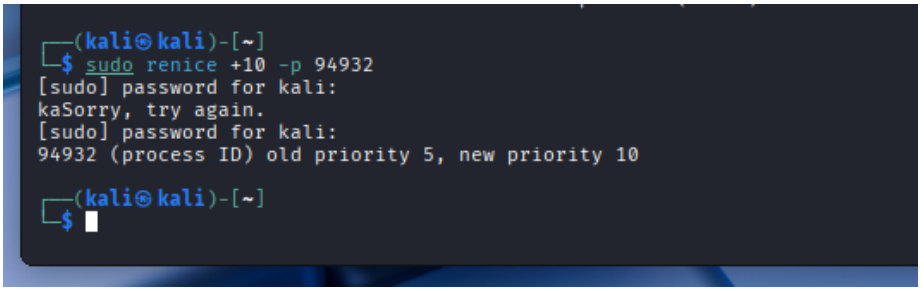
Question 7

Use `pstree` command with option that show the number of threads to each process.

Command
pstree -p $(pgrep -o mainprocess)
Output


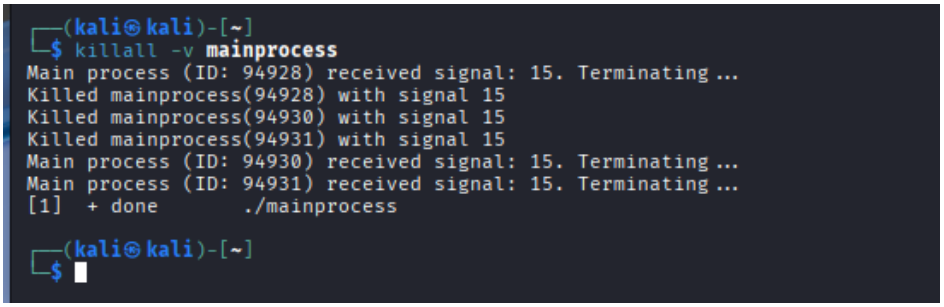
Question 8

Use `renice` command to change priority level of one of process “subprocess1”.

Command
sudo renice +10 -p 94932
Output


Question 9

Terminate all running processes with the name “mainprocess”.

Command
killall -v mainprocess
Output


Question 10

Write a short C or Python code (choose only one language) demonstrating multiprocessing with `fork()` and `wait()`. Compile and/or run the code. Show the output.

Source Code:

```
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>

int main() {
    pid_t pid = fork();

    if (pid == 0) {
        // Child process
        printf("I am the Child process (PID: %d)\n", getpid());
    } else {
        // Parent process
        wait(NULL);
        printf("I am the Parent process (PID: %d). My child has finished.\n", getpid());
    }
    return 0;
}
```

Output:

